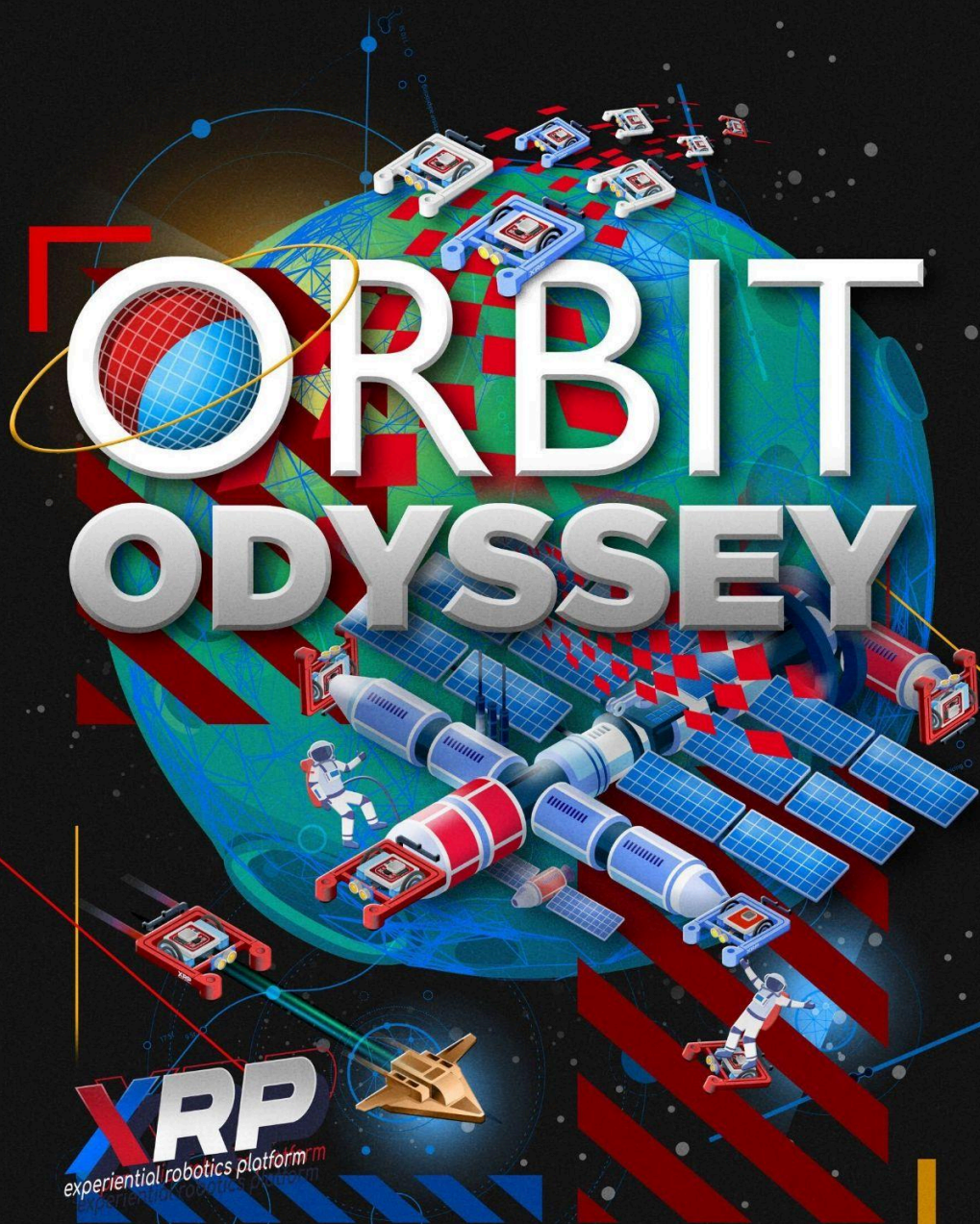


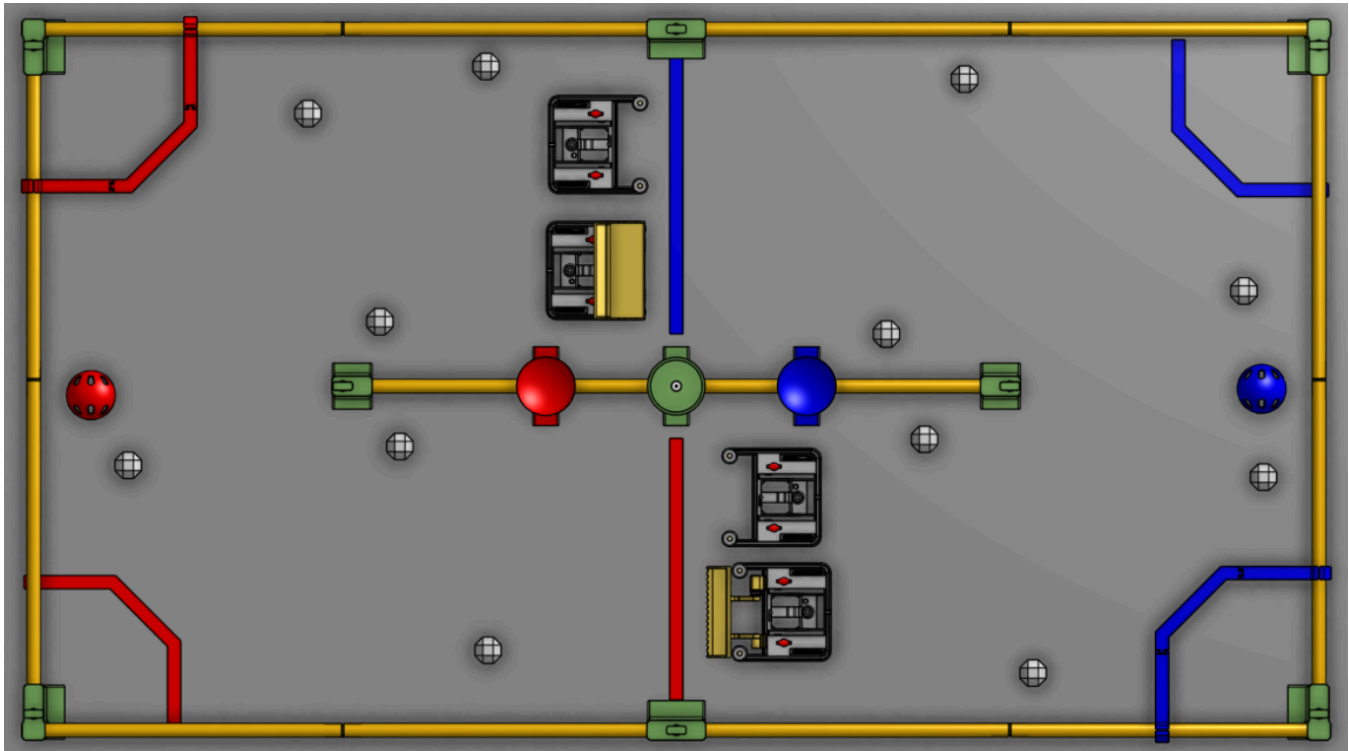


XRP
experiential robotics platform

experientialrobotics.org

Game Manual

1. Game Overview

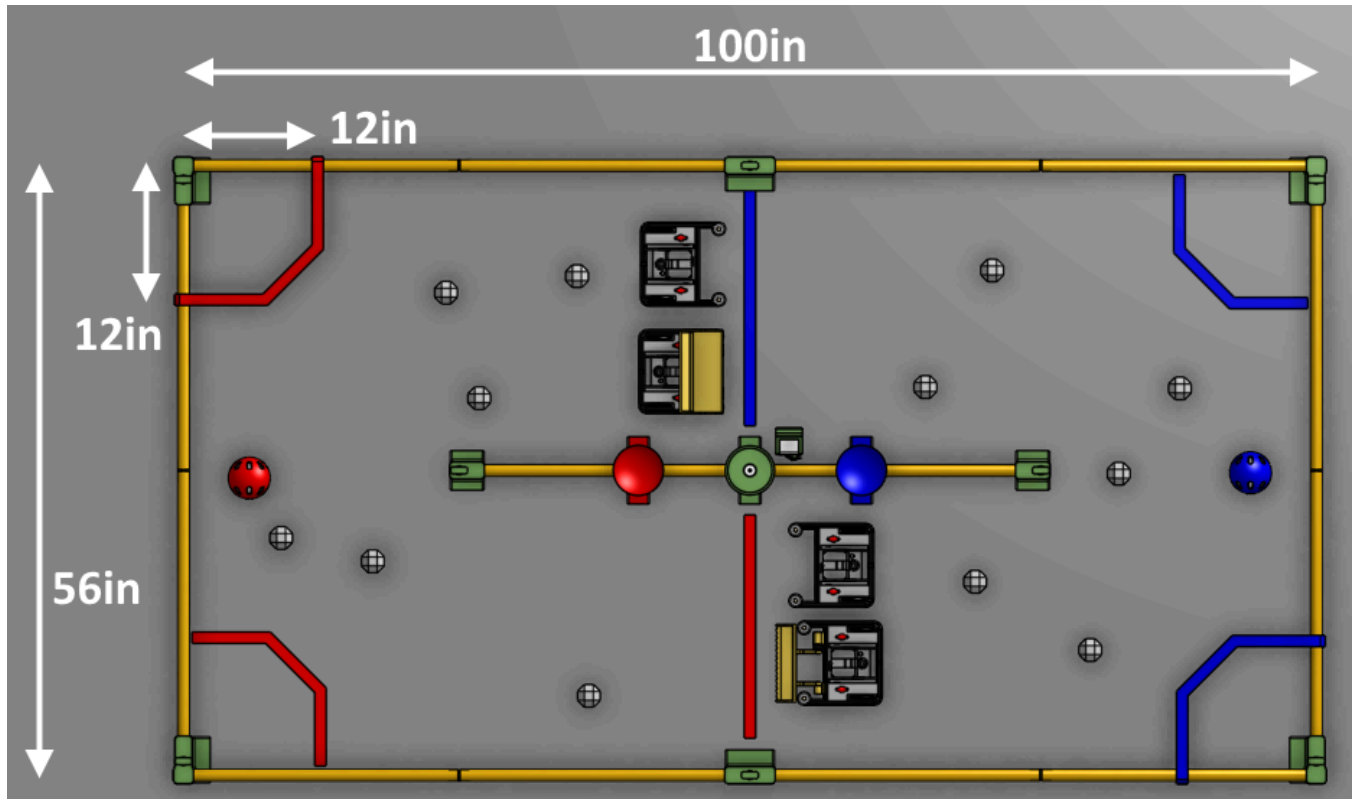


1. The challenge will begin with a 30-second autonomous period. (See starting positions above.)
 - 1.1. Robots can score points by navigating to the *Low Rubble Zone* or *High Rubble Zone*, scoring preloaded *Rubble*. Each piece of *Rubble* scored is worth 1 point in the *Low Rubble Zone* and 2 points in the *High Level Zone*.
 - 1.2. Robots that position themselves at the end of the autonomous period (even partially) in the *Low Rubble Zone* score an additional 5 points.
2. After 30 seconds, teams will be given a transition period to convert to driver-controlled.
3. The 2-minute driver-controlled period of the challenge will begin after teams have repositioned their robots on the playing field after driver-controlled programming.
 - 3.1. During the driver-controlled period, robots should race around the earth to gather data and collect space rubble. Robots may carry an *Amplifier* with them to gather more data, earning bonus points.
 - 3.1.1. For each *Lap* a robot completes, it will earn 5 points
 - 3.1.2. For each *Lap* a robot completes while holding an *Amplifier*, that robot earns 3 extra points
4. The last 30 seconds is the endgame.
 - 4.1. An alliance will earn 10 points if its amplifier is returned to earth (on the pedestal at the end of the match)
 - 4.2. Robots may also earn points for parking (even partially) in the cleared *Low Rubble Zone*, worth 5 points.

- 4.3. If all both *Amplifiers* are returned to *Earth* there will be an additional 10 *Coopertition* points added to each team's score.

Autonomous		Points earned
<i>Rubble</i> in alliance <i>Low Rubble Zone</i>		1
<i>Rubble</i> in alliance <i>High Rubble Zone</i>		3
<i>Parking</i> on the <i>Low Rubble Zone</i> of your alliance color		5
Driver controlled		
<i>Robot Orbit Completion</i>		5
<i>Robot Orbit with Amplifier</i>		3
Endgame		
<i>Amplifier</i> returned to <i>Earth</i>		10
<i>Low Rubble Zone Parking</i>		5
<i>Coopertition Bonus</i>		10

2 Field Overview



The Orbit Odyssey field contains three scoring locations.

1. *Rubble Zones*
 - 1.1. Each alliance has a *Low and High Rubble Zone*.
 - 1.1.1. The *High Rubble Zone* is 12 in. by 12 in. and 2 in. tall.
 - 1.1.2. The *Low Rubble Zone* is 12 in. x 12 in.
2. *Earth Pedestal*
 - 2.1. The Earth Pedestal is 5 in. tall
3. *Centerline*
 - 3.1. The *Centerline* divides the field. Robots may not cross the *Centerline* (except for on top of the *Pass*) until after the driver-controlled period begins.
4. *Orbit*
 - 4.1. A *Orbit* is defined as robot moving clockwise around the field 360 degrees
5. *Game Pieces*
 - 5.1. *Rubble*
 - 5.1.1. Each *Rubble* is a white 3D print, with a nominal diameter of 2 in.
 - 5.2. *Amplifier*
 - 5.2.1. Each *Amplifier* is an alliance specific colored wiffle ball with a diameter of 3.5 inches.

3 Robot Construction Rules

1. Each robot must follow these rules in order to pass inspection and compete.
 - 1.1. A robot in its starting configuration must fit inside an 10 in. cube.
 - 1.2. A robot can weigh no more than 1 kg.
2. A robot may not contain materials or components that could cause damage to the playing field.
3. A robot should have a method of safely storing energy; for example, no launching rubber bands.
4. Other robot construction rules may be determined by your facilitator.
5. Robots must prominently display their **alliance color** and **team number** during matches.
 - 5.1. The team marker must be 2 in. x 2 in.
 - 5.2. It should have your assigned team number.
 - 5.3. It should be mounted in the groove of the XRP chassis.
 - 5.4. It should be stable enough to not fall out in the competition.
 - 5.5. You should have a red and blue version of the marker.

4 Penalty

Breaching the rules will result in a penalty or a red card.

1. Ticket (2 points to the offended alliance)
 - 1.1. Pinning a robot for 5 or more seconds
 - 1.2. During the autonomous period:
 - 1.2.1. Drive teams may not interact with the robot or operator controls (except for a single brief button press).
 - 1.2.2. Robots should remain on the side of the field with their corresponding *Rubble Zones*.
 - 1.3. Robots cannot enter the other alliance's *Rubble Zones*.
 - 1.4. A robot may not damage or functionally impair another robot or the playing field.
 - 1.5. Robots may possess only 1 *Rubble* at a time (1 penalty per *Rubble*).
 - 1.6. Robots may not intentionally detach or leave parts on the field
 - 1.7. A robot may not launch, shoot, or throw a game piece
 - 1.8. No one can enter the field during a match.
2. Red card (Disqualification from the match)
 3. Excessive, ungracious, or unprofessional behavior will result in a red card.
 4. Throwing *Rubble* or *Amplifiers* at anyone is considered an automatic red card.
 5. The team must send at least 1 team member to a match whether the robot is there or not.
 6. Intentionally violating the rules may result in a red card.

5 Match Play

Match setup

- 1.1. During match setup, teams put their robots in their starting positions and preload *Rubble* if desired.

- 1.1.1. **Please make sure to avoid stepping on the field elements while putting robots onto the field.**
- 1.2. Starting positions
 - 1.2.1. Robots begin the match with all mechanisms within their frame perimeter and must be on their side of the field.
 - 1.2.2. Robots must start with the chassis extending over the centerline facing their corresponding *Rubble Zones*.

Autonomous period

- 1.3. Each match begins with a 30-second autonomous period.
 - 1.3.1. A sound will play to signal that the autonomous period has started.
 - 1.3.2. At the start of the autonomous period, each driver may briefly press one button or key to signal to their robot that the autonomous period has started.
 - 1.3.3. After that, drivers may not touch their controls until the start of the driver-controlled period.
- 1.4. Teams will be given time to change operation modes between autonomous and driver controlled.

Driver-controlled period

- 1.5. Once the autonomous period ends, the 2-minute driver-controlled period begins.
- 1.6. A sound will play to signal that the autonomous period has ended and the driver-controlled period has begun.
- 1.7. During the driver-controlled period, drivers are free to touch their controls.
- 1.8. Once there are thirty seconds left in the driver-controlled period, a sound will play to signal that the endgame phase has begun.
- 1.9. A sound will play to signal that the driver-controlled period has ended.
- 1.10. Once the driver-controlled period has ended, drivers must stop providing input to their robots.

6 Competition Setup

The competition will consist of randomly selected 2 versus 2 matches. The team with the highest average score after the last match is declared the event winner. Awards will also be given based on programming, creativity, and *Gracious Professionalism*.

Use ChatGPT to create your own match schedule. See the prompt below and results for 4 teams with a 2-on-2 alliance.

You

Can you create a match schedule for a 2 on 2 tournament with 4 teams with a total of 8 matches

ChatGPT

Sure, here's a match schedule for a 2 on 2 tournament with 4 teams and a total of 8 matches:

Match 1: Team 1 vs. Team 2

Match 2: Team 3 vs. Team 4

Match 3: Team 1 vs. Team 3

Match 4: Team 2 vs. Team 4

Match 5: Team 1 vs. Team 4

Match 6: Team 2 vs. Team 3

Match 7: Team 1 vs. Team 3

Match 8: Team 2 vs. Team 4

Each team plays against each other e, resulting in 6 matches. Then, to reach a total of 8 matches, some teams will have to play each other again.

Use the following as a scorecard and tracking sheet to track each match and team's score.

7 Scorecards

Teams			Teams		
Match #			Match #		
Autonomous	Points earned	Total	Autonomous	Points earned	Total
Rubble in alliance Low Rubble Zone	1		Rubble in alliance Low Rubble Zone	1	
Rubble in alliance High Rubble Zone	3		Rubble in alliance High Rubble Zone	3	
Parking on the Low Rubble Zone	5		Parking on the Low Rubble Zone	5	
Driver controlled			Driver controlled		
Robot Lap Completion	5		Robot Lap Completion	5	
Robot Lap with Amplifier	3		Robot Lap with Amplifier	3	
Endgame			Endgame		
Amplifier returned to Earth	10		Amplifier returned to Earth	10	
Low Rubble zone Parking	5		Low Rubble zone Parking	5	
Coopertition Bonus	10		Coopertition Bonus	10	

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